

Project 2 Phase 1: Tag Detection and Pose Estimation

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1 Introduction

This report presents the implementation and results for Project 2 Phase 1: Tag Detection and Pose Estimation. The objective is to estimate the camera's pose (rotation matrix R and translation vector t) relative to a known ARuco tag marker using the Perspective-n-Point (PnP) algorithm with the Direct Linear Transformation (DLT) method and refinement using Gauss-Newton.

2 Methodology

Step 1: Linear Solution (DLT)

1. **Construct Matrix A :** Use $N \geq 4$ pairs of 3D-2D point correspondences.
2. **SVD Solving:** Solve $Ax = 0$ via Singular Value Decomposition to obtain the homography matrix H .
3. **Extract Pose:** Compute $K^{-1}H = [\tilde{h}_1 \quad \tilde{h}_2 \quad \tilde{h}_3]$.
4. **Enforce Orthogonal Constraints:**
 - Perform SVD on $[\tilde{h}_1 \quad \tilde{h}_2 \quad \tilde{h}_1 \times \tilde{h}_2]$ to yield U and V .
 - Compute refined rotation: $R = UV^T$.
 - Compute translation scale: $t = \tilde{h}_3 / \|\tilde{h}_1\|$.

Step 2: Non-linear Optimization (Gauss-Newton)

1. **Compute Residuals:** $\gamma_i = \begin{bmatrix} u_i \\ v_i \end{bmatrix} - \pi(K(RX_{w,i} + t))$.
2. **Compute Jacobians:** $J_i = \frac{\partial \gamma_i}{\partial [\theta, t]}$ (2×6 matrix).
3. **Construct Normal Equations:** $A = \sum J_i^T J_i$ and $b = -\sum J_i^T \gamma_i$.
4. **Solve Increments:** $\begin{bmatrix} \delta\theta \\ \delta t \end{bmatrix} = A^{-1}b$.
5. **Update State:** $\begin{bmatrix} \theta \\ t \end{bmatrix} = \begin{bmatrix} \theta_0 \\ t_0 \end{bmatrix} + \begin{bmatrix} \delta\theta \\ \delta t \end{bmatrix}$.
6. **Iterate:** Repeat steps 1-5 until convergence.

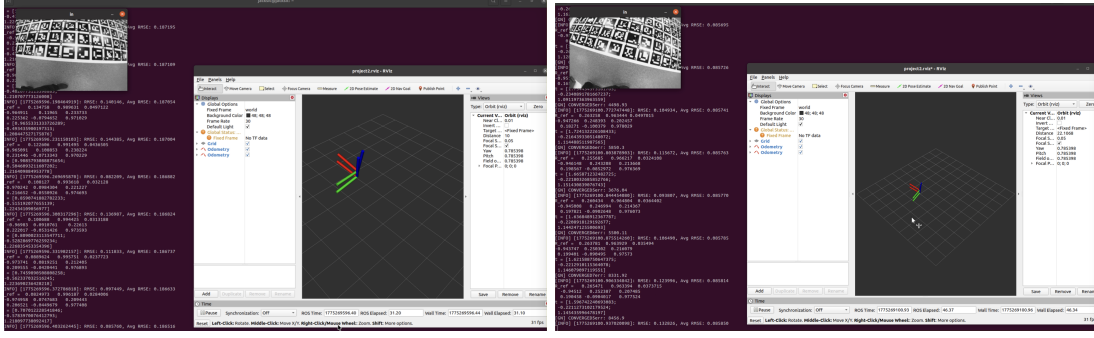


Figure 1: Left: DLT initialization result. Right: Refined pose after Gauss-Newton optimization.

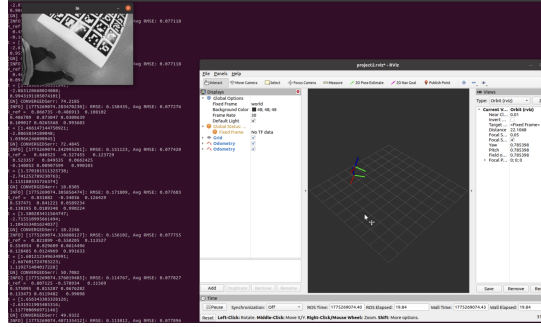


Figure 2: Behavior when encountering a large error during tracking.

3 Results

3.1 Pose Estimation Visualization

3.2 Statistics

The tracking demonstrated high accuracy. Across the recorded sequence, the average Root Mean Square Error (RMSE) for the initial Direct Linear Transformation (DLT) estimation was **0.237696** pixels. After applying the Gauss-Newton nonlinear refinement, the average RMSE significantly improved to **0.092853** pixels.

4 Analysis

Accuracy: The DLT method provided a good initial guess, but the Gauss-Newton nonlinear refinement significantly improved the accuracy by minimizing the exact geometric reprojection error.

Failure Cases: The pose estimation can become unstable when too few points are visible.

5 References

1. ELEC 5660 Course Materials and Slides.
2. Gao Xiang, "Visual SLAM 14 Lectures": https://github.com/lishuwei0424/Self_Reference_Book_for_VSALM/blob/master/13.%E9%AB%98%E7%BF%94%20%E8%A7%86%E8%A7%89SLAM%E5%8D%81%E5%9B%9B%E8%AE%B2-%E5%AE%8C%E6%95%B414%E8%AE%B2%E7%89%88.pdf